

SANYA PHOENIX INTL, China

WMO: 574941

Lat: 18.303N

Lon: 109.412E

Elev: 92

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	58.9	61.0	46.0	46.1	64.3	49.9	53.4	65.8	19.8	78.7	18.6	77.4	5.6	80	0.355

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	8.8	91.6	81.3	90.0	80.8	89.7	80.7	83.5	88.8	82.7	87.8	82.2	87.3	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.2	168.4	88.2	80.9	161.3	87.3	80.7	160.0	87.2	47.5	88.7	46.9	89.4	46.4	87.9	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.0	14.1	DB	53.0	93.6	3.0	1.6	50.8	94.7	49.1	95.6	47.4	96.5	45.2	97.6	
			WB	47.8	84.3	2.6	0.9	45.9	85.0	44.4	85.5	43.0	86.1	41.1	86.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.5	72.1	73.6	77.0	81.1	84.0	84.8	83.5	83.2	82.3	80.3	78.0	73.8
	DBStd	5.27	4.17	4.30	3.80	2.43	2.37	2.28	2.34	2.07	1.96	2.35	3.13	4.24
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	12	4	4	1	0	0	0	0	0	0	0	0	3
	CDD50	10764	686	660	837	934	1055	1043	1038	1029	968	938	839	737
	CDD65	5300	225	243	373	484	590	593	573	564	518	473	389	275
	CDH74	53303	964	1296	2745	4967	7321	7661	6965	6618	5711	4436	3120	1499
	CDH80	18556	104	169	535	1520	3114	3463	2787	2480	1988	1341	798	257
Wind	WSAvg	6.6	7.7	7.3	7.3	6.6	5.7	5.4	5.8	5.9	5.8	7.0	7.3	7.7
Precipitation	PrecAvg	69.6	0.7	0.7	1.2	3.1	6.8	7.2	9.3	10.4	12.3	12.1	4.2	1.7
	PrecMax	85.6	2.8	3.1	4.1	7.1	15.7	14.6	15.5	20.6	23.5	36.5	12.8	7.3
	PrecMin	46.9	0.0	0.1	0.1	0.1	1.6	2.8	5.0	1.6	4.9	1.5	0.2	0.1
	PrecStd	10.7	0.6	0.7	1.1	1.8	3.3	2.7	2.7	3.8	5.5	7.8	3.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.6	85.6	87.4	90.0	92.9	92.9	91.8	91.7	91.3	91.3	89.3	86.4
		MCWB	71.8	73.4	75.5	78.9	81.1	81.8	81.5	81.8	80.2	77.0	76.0	74.3
	2%	DB	82.6	83.9	85.7	88.1	91.0	91.5	90.0	89.9	89.7	89.4	87.7	84.5
		MCWB	71.5	73.5	75.8	78.7	81.1	81.9	81.4	81.4	80.5	76.9	75.9	72.9
	5%	DB	80.7	82.3	84.3	87.4	89.7	90.0	89.5	89.3	88.2	87.8	86.0	82.7
		MCWB	70.8	72.9	75.4	78.6	80.6	81.4	81.2	81.2	80.2	76.8	75.1	71.9
	10%	DB	78.9	80.6	82.7	86.0	88.2	89.5	88.1	87.9	87.7	86.2	84.3	80.8
		MCWB	69.4	72.0	74.4	78.0	80.5	81.2	80.8	80.7	80.1	76.3	74.4	70.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	76.3	78.3	81.0	83.9	84.4	84.2	84.1	83.0	80.7	79.6	76.3
		MCDB	80.4	82.3	84.6	86.4	89.1	89.7	89.6	89.4	87.9	87.0	86.4	84.0
	2%	WB	74.1	75.1	77.3	80.1	82.8	83.5	82.8	82.7	82.2	79.7	78.1	74.8
		MCDB	79.3	80.9	83.6	85.8	87.8	88.8	87.9	87.8	87.1	85.6	84.6	81.9
	5%	WB	72.5	74.2	76.4	79.6	82.1	82.7	82.2	81.8	81.1	78.7	77.1	73.6
		MCDB	78.3	80.0	82.6	85.3	87.2	87.9	87.3	87.0	86.5	84.3	83.3	80.1
	10%	WB	71.3	73.2	75.6	78.8	81.4	82.1	81.4	81.0	80.4	77.9	76.1	72.3
		MCDB	77.4	78.9	81.6	84.5	86.8	87.3	86.6	86.3	85.8	83.4	82.0	78.9
Mean Daily Temperature Range		MDBR	13.1	12.0	11.0	10.0	9.2	8.8	8.8	9.3	10.4	12.7	13.3	13.6
	5% DB	MCDBR	13.7	12.9	11.1	10.7	9.6	9.1	9.8	10.6	11.6	14.6	14.4	14.8
		MCWBR	6.4	5.9	5.1	4.9	4.7	4.8	5.3	5.4	5.8	6.2	6.4	6.7
	5% WB	MCDBR	11.6	10.7	9.7	9.2	8.5	8.5	8.9	9.7	10.5	11.4	12.1	13.5
		MCWBR	5.9	5.5	5.0	4.9	4.7	4.9	5.2	5.5	5.7	5.6	6.1	6.7
Clear-Sky Solar Irradiance	taub	0.505	0.525	0.629	0.593	0.484	0.470	0.451	0.477	0.532	0.569	0.510	0.529	
	taud	2.022	2.006	1.781	1.921	2.275	2.329	2.379	2.291	2.098	1.941	2.079	1.966	
	Ebn at Noon	245	248	227	237	261	262	267	263	248	233	242	233	
	Edh at Noon	52	56	72	63	43	41	39	43	51	58	49	54	
All-Sky Solar Radiation	RadAvg	1164	1358	1479	1735	1833	1721	1694	1611	1517	1427	1280	1098	
	RadStd	145	189	112	136	125	142	163	114	157	172	104	109	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (5 neighbors)	+0.47	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	+162	+149

Nomenclature: See separate page